Lecture 14

Modelling Software Systems

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- Models of a software system's design are:
  1. Abstract representations of knowledge of the system.
  2. An approximation of some aspects of the system.
  3. Can be expressed using graphs, natural language, math, and other ways.
- A system's design models are used to:
  1. Communicate knowledge about the system.
  2. Help articulate and assess design decisions for the system by abstracting away certain details.
  3. Act as documentation to aid in maintaining the system.
- Design models can be 'formal' or 'rigorous'.
  - Don't let the word rigorous confuse you! Rigorous design models are subject to interpretability.
- Examples of formal design models include:
  1. FSMs.
  2. Graphs.
  3. Relations.
- Examples of rigorous design include:
  1. UML diagrams.
  2. Proprietary/non-standard models.
- Analysis class diagrams are an example of a rigorous design model.
- Analysis class diagrams are composed of three types of classes:
  1. Boundary classes.
     - Are interfaces between the system and its environment
     - Keep the format of communication between the system and its environment as its secret.
  2. Entity classes.
     - Store data.
     - Exchange data between classes.
     - Keep an n-tuple (representing the data the entity class stores) as its secret.
  3. Control classes.
     - Coordinate interactions between other classes in response to a business event.
- The $4 + 1$ model is another example of design modelling.
  - The scenario view of the system acts as the basis for its logical view.
    * The logical view represents the system as an analysis class diagram.

        · All other views are derived from the logical view.

- All software architectures can be said to be composed of the following parts:
  1. Elements.
  2. Connectors.
  3. The rationale behind those elements/connectors.
- Connectors always have some source and destination element.
- 'Multiplicity' is an attribute of connectors.
  - The multiplicity of a connector describes how many elements are related together by the connector from source to destination.
    * The multiplicity of any connector is one of:
      1. 1-1.
      2. 1-many.
      3. Many-1.
      4. Many-many.
- 'Distance' is another attribute of a connector.
  - The distance of a connector describes how 'far apart' the elements it connects are.
    * How far apart elements are can be categorized as:
      1. Elements belonging to the same thread.
      2. Elements belonging to the same process.
      3. Elements belonging to the same computer.
      4. Etc.